

Agentic AI Testing - Complete Test Plan

1. Overview

This document defines a complete testing strategy for Agentic AI systems, including unit, integration, security, performance, end-to-end, and user acceptance testing.

2. Requirements

- 1 Agent should correctly select tools
- 2 Handle failures and retries
- 3 Maintain response quality
- 4 Respect cost and latency constraints
- 5 Ensure security and data privacy

3. Unit Test Cases

- 1 Validate tool schema parsing
- 2 Test prompt formatting logic
- 3 Test retry logic for failures
- 4 Test memory storage and retrieval

4. Integration Test Cases

- 1 Agent + LLM integration
- 2 Agent + tool APIs
- 3 RAG pipeline (retrieval + generation)
- 4 Multi-tool chaining workflows

5. Security Test Cases

- 1 Prompt injection attacks
- 2 Data leakage validation
- 3 Unauthorized API access
- 4 Input sanitization tests

6. Performance Test Cases

- 1 Latency under load
- 2 Throughput testing
- 3 Cost per request monitoring
- 4 Scaling with concurrent users

7. End-to-End Test Cases

- 1 Full workflow execution
- 2 Failure recovery scenarios
- 3 Multi-step reasoning validation
- 4 Real tool + real LLM validation

8. UAT Test Cases

- 1 User satisfaction
- 2 Response correctness
- 3 Ease of interaction
- 4 Business goal alignment

9. Test Strategy Layers

- 1 Layer 1: Tool contract tests
- 2 Layer 2: Agent logic tests
- 3 Layer 3: Evaluation tests
- 4 Layer 4: End-to-end tests

10. Metrics

- 1 Accuracy
- 2 Hallucination rate
- 3 Latency
- 4 Cost
- 5 Success rate